



Sales & Data Analysis Report

JAN, FEB, MAR 2019



Capstone Project

Given dataset contains sales transactions from three different branches of Amazon, respectively located in Mandalay, Yangon and Naypyitaw located in Japan . The data contains 17 columns and 1000 rows



Approach Used

1.Data Wrangling:

1.1 Build a database

```
create table Amazon(  
Invoice_ID VARCHAR(30),  
Branch VARCHAR(5),  
City VARCHAR(30),  
Customer_type VARCHAR(30),  
Gender VARCHAR(10),  
Product_line VARCHAR(100),  
Unit_price DECIMAL(10,2),  
Quantity INT,  
VAT FLOAT(6,4),  
Total DECIMAL(10,2),  
Date DATE,  
Time TIMESTAMP,  
Payment DECIMAL(10,2),  
cogs DECIMAL(10,2),  
gross_margin_percentage  
FLOAT(11,9),  
gross_income DECIMAL(10,2),  
Rating FLOAT(2,1) );
```

1.2 insert the data.

```
LOAD DATA INFILE  
'/Users/mayan/Downloads/Am  
azon.csv'  
INTO TABLE amazon  
FIELDS TERMINATED BY ','  
ENCLOSED by '"'  
LINES TERMINATED BY '\n'  
IGNORE 1 ROWS;
```

2.Feature Engineering:



2.1 Add a new column named timeofday

```
Alter table amazon
add column timeofday
varchar(10);

update amazon
set timeofday = case
  when extract(hour from Time)
  >= 6 and extract(hour from
Time) < 12 then "morning"
  when extract(hour from Time)
  >= 12 and extract(hour from
Time) < 18 then "afternoon"
  else "evening"
end;
```

2.2 Add a new column named dayname

```
alter table amazon
add column dayname
varchar(10);

update amazon
set dayname = case
weekday(date)
when 0 then 'mon'
when 1 then 'tue'
when 2 then 'wed'
when 3 then 'thu'
when 4 then 'fri'
when 5 then 'sat'
when 6 then 'sun'
end;
```

2.3 Add a new column named monthname

```
alter table amazon
add column monthname varchar(10);

update amazon
set monthname = case
  when MONTHNAME(Date) = 'january' then 'jan'
  when MONTHNAME(Date) = 'february' then 'feb'
  when MONTHNAME(Date) = 'march' then 'mar'
  when MONTHNAME(Date) = 'april' then 'apr'
  when MONTHNAME(Date) = 'may' then 'may'
  when MONTHNAME(Date) = 'june' then 'jun'
  when MONTHNAME(Date) = 'july' then 'jul'
  when MONTHNAME(Date) = 'august' then 'aug'
  when MONTHNAME(Date) = 'september' then 'sep'
  when MONTHNAME(Date) = 'october' then 'oct'
  when MONTHNAME(Date) = 'november' then 'nov'
  when MONTHNAME(Date) = 'december' then 'dec'
end;
```

Business Questions To Answer:



3. Exploratory Data Analysis (EDA):

1.What is the count of distinct cities in the dataset?

2.For each branch, what is the corresponding city?

3- What is the count of distinct product lines in the dataset?

select count(distinct city) as
distinct_city_Count from
amazon;

	distinct_city_Count
▶	3

select branch, city
from amazon
group by city, branch;

	branch	city
▶	A	Yangon
	C	Naypyitaw
	B	Mandalay

select count(distinct Product_line) as
distinct_product_lines_count
from amazon;

	distinct_product_lines_count
▶	6



4-Which payment method occurs most frequently?

```
select payment ,
count(*) as occurrence
from amazon
group by payment
order by occurrence desc
limit 1;
```

	payment	occurrence
▶	Ewallet	345

5- Which product line has the highest sales?

```
select product_line,
sum(unit_price * quantity) as
total_sales
from amazon
group by Product_line
order by total_sales desc
limit 1;
```

	product_line	total_sales
▶	Food and beverages	53471.280000000006

6- How much revenue is generated each month?

```
select date_format(date , '%y-%m') as month,
sum(unit_price * quantity) as total_revenue
from amazon
group by date_format(date,'%y-%m')
order by month asc;
```

	month	total_revenue
▶	19-01	110754.16
	19-02	92589.88
	19-03	104243.33999999997



7- In which month did the cost of goods sold reach its peak?

8- Which product line generated the highest revenue?

9- In which city was the highest revenue recorded?

```
select date_format(date,'%y-%m') as month,
sum(cogs * quantity) as total_cogs
from amazon
group by date_format(date,'%y-%m')
order by total_cogs desc
limit 1;
```

	month	total_cogs
▶	19-01	795133.68000000005

```
select product_line,
sum(unit_price * quantity) as total_revenue
from amazon
group by Product_line
order by total_revenue desc
limit 1;
```

	product_line	total_revenue
▶	Food and beverages	53471.280000000006

```
select city, sum(unit_price * quantity) as total_revenue
from amazon
group by city
order by total_revenue desc
limit 1;
```

	city	total_revenue
▶	Naypyitaw	105303.53



- 10- Which product line incurred the highest Value Added Tax?
- 11- For each product line, add a column indicating "Good" if its sales are above average, otherwise "Bad."
- 12- Identify the branch that exceeded the average number of products sold.

```
select product_line, sum(VAT) as
total_VAT
from amazon
group by Product_line
order by total_VAT desc
limit 1;
```

product_line	total_VAT
Food and beverages	2673.5639999999999

```
alter table amazon
add column sales_category varchar(10);

with product_line_sales as(select product_line, sum(unit_price
* quantity) as total_sales
from amazon
group by product_line),
average_sales as(select avg(total_sales) as avg_sales from
product_line_sales)
update amazon
set sales_category = case
when product_line in (select product_line from
product_line_sales, average_sales
where product_line_sales.total_sales >
average_sales.avg_sales)
then 'good'
else 'bad'
end;
```

```
select branch, sum(quantity)
as total_quantity
from amazon
group by branch
having sum(Quantity) > (
select avg(total_quantity)
from (select sum(quantity)
as total_quantity
from amazon
group by branch) as
subquery);
```

	branch	total_quantity
▶	A	1859



13- Which product line is most frequently associated with each gender?

```
select gender, product_line, count(*)
as frequency
from amazon
group by gender,product_line
order by gender, frequency desc;
```

	gender	product_line	frequency
►	Female	Fashion accessories	96
	Female	Food and beverages	90
	Female	Sports and travel	88
	Female	Electronic accessories	84
	Female	Home and lifestyle	79
	Female	Health and beauty	64
	Male	Health and beauty	88
	Male	Electronic accessories	86
	Male	Food and beverages	84
	Male	Fashion accessories	82
	Male	Home and lifestyle	81
	Male	Sports and travel	78

14- Calculate the average rating for each product line.

```
select product_line, avg(rating) as
average_rating
from amazon
group by Product_line;
```

	product_line	average_rating
►	Health and beauty	7.003289473684212
	Electronic accessories	6.92470588235294
	Home and lifestyle	6.8375
	Sports and travel	6.916265060240964
	Food and beverages	7.113218390804598
	Fashion accessories	7.029213483146067

15- Count the sales occurrences for each time of day on every weekday.

```
select dayofweek(date) as day_of_week,
hour(time) as hour_of_day,
count(*) as sales_occurrences
from amazon
where dayofweek(date) in (1, 7)
group by day_of_week, hour_of_day
order by day_of_week, hour_of_day;
```

lay_of_week	hour_of_day	sales_occurre
	14	13
	15	11
	16	6
	17	11
	18	15
	19	17
	20	9
	10	16
	11	12
	12	18
	13	14
	14	11
	15	12



16- Identify the customer type contributing the highest revenue.

```
select customer_type,
sum(unit_price * quantity) as
total_revenue
from amazon
group by Customer_type
order by total_revenue desc
limit 1;
```

	customer_type	total_revenue
▶	Member	156403.279999999985

17- Determine the city with the highest VAT percentage.

```
select city, max(VAT) as
highest_VAT_percentage
from amazon
group by city
order by highest_VAT_percentage desc
limit 1;
```

	city	highest_VAT_percentage
▶	Naypyitaw	49.65

18- Identify the customer type with the highest VAT payments.

```
select customer_type, sum(unit_price *
quantity * VAT / 100) as total_VAT_paid
from amazon
group by Customer_type
order by total_VAT_paid desc
limit 1;
```

	customer_type	total_VAT_paid
▶	Member	38358.945572200064



19- What is the count of distinct customer types in the dataset?

```
select count(distinct customer_type)
as distinct_customer_types
from amazon;

select distinct customer_type from
amazon;
```

	customer_type
▶	Member
	Normal
	Normal

20- What is the count of distinct payment methods in the dataset?

```
select count(distinct payment) as
distinct_payment_methods
from amazon;

select distinct payment from amazon;
```

	payment
▶	Ewallet
	Cash
	Credit card

21- Which customer type occurs most frequently?

```
select customer_type, count(*) as frequency
from amazon
group by Customer_type
order by frequency desc
limit 1;
```

	customer_type	frequency
▶	Member	501



22- Identify the customer type with the highest purchase frequency.

```
select customer_type,
count(*) as purchase_frequency
from amazon
group by Customer_type
order by purchase_frequency desc
limit 1;
```

Result Grid			Filter Rows:
	customer_type	purchase_frequency	
▶	Member	501	

23- Determine the predominant gender among customers.

```
select gender, count(*) as
gender_count
from amazon
group by gender
order by gender_count desc
limit 1;
```

	gender	gender_count
▶	Female	501

24- Examine the distribution of genders within each branch.

```
select branch, gender, count(*) as
gender_count
from amazon
group by branch, gender
order by branch, gender_count desc;
```

	branch	gender	gender_count
▶	A	Male	179
	A	Female	161
	B	Male	170
	B	Female	162
	C	Female	178
	C	Male	150



25- Identify the time of day when customers provide the most ratings.

```
select date(date) as order_date,
hour(time) as hour_of_day, count(*)
as ratings_count
from amazon
where rating is not null
group by order_date, hour_of_day
order by ratings_count desc
limit 1 ;
```

	order_date	hour_of_day	ratings_count
▶	2019-03-20	19	5

26- Determine the time of day with the highest customer ratings for each branch.

```
with Rankedratings as (
select branch, date(Date) as
order_date, hour(time) as hour_of_day,
count(*) as rating_count,
row_number() over(partition by
branch, date(Date) order by count(*)
desc) as Ranks
from amazon
where rating is not null
group by branch, order_date,
hour_of_day )
select branch, order_date,
hour_of_day, rating_count
from Rankedratings
where ranks = 1
order by branch, order_date,
rating_count desc;
```

27- Identify the day of the week with the highest average ratings.

```
select dayofweek(date) as day_of_week,
avg(rating) as average_rating
from amazon
where rating is not null
group by day_of_week
order by average_rating desc
limit 1;
```

	day_of_week	average_rating
▶	2	7.153599999999999



28- Determine the day of the week with the highest average ratings for each branch.

```
with AverageRatings as(
select branch, dayofweek(date) as
day_of_week, avg(rating) as average_rating
from amazon
where rating is not null
group by branch, day_of_week )
select branch, day_of_week, average_rating
from AverageRatings
where (branch, average_rating) in (
select branch, max(average_rating) as
highest_average
from AverageRatings
group by branch )
order by branch, average_rating desc;
```

	branch	day_of_week	average_rating
▶	A	6	7.31199999999999985
	B	2	7.335897435897434
	C	6	7.278947368421051



Thank you